

Question	Answer	Marks
1	C	1

Question	Answer	Marks
4	One mark for each point max three. - variables / constants are used to store items of data - the data stored in variables / constants are accessed by an identifier // named data stores - the value of a variable may change during the execution of a program - the value of a constant will remain the same during the execution of a program	3

Question	Answer	Marks
10(a)	One mark for each correct line DECLARE X : INTEGER DECLARE Y : REAL DECLARE Z : BOOLEAN	3
10(b)	One mark for using FUNCTION and ENDFUNCTION and RETURNS BOOLEAN One mark for naming the function Same One mark for defining the two parameters correctly One mark for comparing the two parameters using ROUND One mark for correctly returning TRUE and FALSE One mark for correct function call Example definition: <pre>FUNCTION Same (A : INTEGER, B : REAL) RETURNS BOOLEAN IF A = ROUND(B, 0) THEN RETURN TRUE ELSE RETURN FALSE ENDIF ENDFUNCTION</pre> Example call: Z ← Same (X, Y)	6
10(c)	A function is defined once and called many times or Define – setting up the function and call is using a function	1

Question	Answer	Marks
2	A	1

Question	Answer	Marks
4	One mark per mark point, max four DIV, max two - To perform integer division - Meaning only the whole number part of the answer is retained - Example of DIV For example <code>DIV(9, 4) = 2</code> ROUND, max two - To return a value rounded to a specified number of digits / decimal places - The result will either be rounded to the next highest or the next lowest value ... depending on whether the value of the preceding digit is ≥ 5 or < 5 - Example of ROUND for example, <code>ROUND(4.56, 1) = 4.6</code>	4

Question	Answer	Marks
6	One mark for each correct feature, max two One mark for each correct accompanying reason, max two For example: Meaningful identifiers – to enable the programmer (or future programmers) to easily recognize the purpose of a variable / array / constant // to enable easy tracking of a variable / constant / array through the program Use of comments – to annotate each section of a program so that a programmer can find specific sections / so that the programmer knows the purpose of that section of code Procedures and functions – to make programs modular and easier to update / add functionality	4

Question	Answer	Marks
8(a)	One mark per mark point, max three - Storing string in Phrase - Correct use of LENGTH function - Correct use of UCASE function - Correct outputs of LENGTH and UCASE For example: <pre>Phrase ← "The beginning is the most important part" OUTPUT LENGTH(Phrase) OUTPUT UCASE(Phrase)</pre>	3
8(b)	One mark for each correct line, max two <pre>40 THE BEGINNING IS THE MOST IMPORTANT PART</pre>	2

JUNE 2023 (2)

Question	Answer	Marks
6	One mark for each feature and one mark for corresponding example (max six) - ensuring that all identifiers have meaningful names example using <code>Total</code> to store a running total - using comments to explain how the program works example <code>// all values are zeroed before the next calculation</code> - using procedures and functions for the tasks within a program example <code>CalculateInterest(Deposit, Rate)</code>	6

Question	Answer	Marks
11(a)	One mark for any two correct lines <pre> DECLARE P : STRING P ← "The world" DECLARE Q : CHAR Q ← 'W' </pre>	2

Question	Answer	Marks
11(b)	One mark for each point (max four) - converting <code>P</code> to upper case - finding the length of <code>P</code> - using a loop to check for position of <code>Q</code> - using the string operation <code>substring</code> - storing the loop counter in <code>Position</code> if the value is found For example: <pre> P ← UCASE(P) Counter ← 1 Position ← 0 REPEAT IF SUBSTRING(P, Counter, 1) = Q THEN Position ← Counter ENDIF Counter ← Counter + 1 UNTIL Position <> 0 OR Counter = LENGTH(P) </pre>	4
11(c)	5	1

JUNE 2023 (3)

Question	Answer	Marks
1	B	1

Question	Answer	Marks
2	One mark per mark point, max four MOD, max two - To perform (integer) division when one number is divided by another ... and find the remainder - Allow example e.g. $7 \text{ MOD } 2 = 1$ RANDOM, max two - To generate (pseudo) random numbers ...(usually) within a specified range - Allow example e.g. <code>RANDOM() * 10</code> returns a random number between 0 and 10	4

Question	Answer	Marks
3	One mark per mark point, max three MP1 A call statement is used in order to make use of a function // the function is called using its identifier MP2 Parameters are / may be passed (from the main program) to the function (to be used within the function) MP3 The function performs its task ... MP4 ... and returns a value / values to the main program	3

Question	Answer	Marks
9(a)	DECLARE Saying : STRING	1

Question	Answer	Marks
9(b)	One mark per mark point, max five MP1 input a string into Saying MP2 correct use of <code>OPENFILE</code> to write data MP3 correct use of <code>WRITEFILE</code> to write Saying MP4 correct use of <code>CLOSEFILE</code> MP5 correct use of filename <code>Quotations.txt</code> throughout For example: <pre> INPUT Saying OPENFILE "Quotations.txt" FOR WRITE WRITEFILE "Quotations.txt", Saying CLOSEFILE "Quotations.txt" </pre>	5

NOVEMBER 2023

Question	Answer	Marks
1	C	1

Question	Answer	Marks
2	B	1

Question	Answer	Marks												
3	One mark for each correct line from description to data type <table><thead><tr><th>Description</th><th>Data type</th></tr></thead><tbody><tr><td>a whole number</td><td>BOOLEAN</td></tr><tr><td>a single letter</td><td>CHAR</td></tr><tr><td>a word or phrase</td><td>INTEGER</td></tr><tr><td>a number with two decimal places</td><td>REAL</td></tr><tr><td></td><td>STRING</td></tr></tbody></table>	Description	Data type	a whole number	BOOLEAN	a single letter	CHAR	a word or phrase	INTEGER	a number with two decimal places	REAL		STRING	4
Description	Data type													
a whole number	BOOLEAN													
a single letter	CHAR													
a word or phrase	INTEGER													
a number with two decimal places	REAL													
	STRING													

NOVEMBER 2023 (2)

Question	Answer	Marks
5	One mark per mark point, max three MP1 variables and constants should have meaningful identifiers MP2 ...so that programmers/future programmers are able to understand their purpose MP3 they are both used for data storage MP4 constants store values that never change during the execution of a program // by example MP5 variables contain values that have been calculated within the program / can change during the execution of the program // by example	3

Question	Answer	Marks
7	One mark per mark point, max five MP1 storing string in Quote MP2 correct assignment for Start // sending correct start value MP3 correct assignment for Number // sending correct number of characters MP4 use of SUBSTRING function with first parameter as Quote, or equivalent, plus two other parameters MP5 correct use of LCASE function MP6 two correct outputs For example: Quote ← "Learning Never Exhausts The Mind" Start ← 25 Number ← 8 OUTPUT SUBSTRING(Quote, Start, Number) OUTPUT LCASE(Quote)	5

Question	Answer	Marks
8	One mark per mark point, max four Procedures, max three MP1 to enable the programmer to write a collection of programming statements under a single identifier MP2 to allow modular programs to be created // to allow procedures to be re-used within the program or in other programs MP3 to make program creation faster because procedures can be re-used // to enable different programmers to work on different procedures in the same project MP4 to make programs shorter (than using the repeated code) / using less duplication of code // to make programs easier to maintain due to being shorter. Parameters, max three MP5 to pass values from the main program to a procedure / function MP6 ...so that they can be used in the procedure / function MP7 allow the procedure / function to be re-used with different data.	4

NOVEMBER 2023 (3)

Question	Answer	Marks
2	B	1

QUESTIONS

Question	Answer	Marks
7	one mark for first description one mark for matching difference max four • local variables - scope is a defined block of code/subroutine/procedure/function • global variables – scope is the whole program • local variables - value cannot be changed elsewhere in the program • global variables – value can be changed anywhere in the program	4